

# Teng Xin Lin

<https://github.com/tene04>

tengxinlin04@gmail.com

---

## Objective

Data Science and AI student seeking a technical internship to apply my skills in machine learning, data engineering, and automation. Passionate about leveraging tools like PyTorch, Whisper, and Spark to solve real-world problems. Eager to contribute to innovative teams while expanding my hands-on experience.

---

## Education

**Polytechnic University of Madrid – UPM**

*B.Sc. in Data Science and Artificial Intelligence*

Sep. 2022 – Present

---

## Projects

- **Personal project: Flappy Bird AI Comparison - Neuroevolution vs Deep Q-Learning**

Developed and compared AI agents using neuroevolution and Deep Q-Learning (DQL) in the Flappy Bird environment with PyTorch and Gymnasium.

- DQL agent achieved 7x faster learning and significantly better performance than neuroevolution.
- Used PyTorch, Gymnasium, Flappy-bird-gymnasium, matplotlib and numpy.
- Provided insights into the efficiency of reinforcement learning algorithms for game AI development.

- **Academic project: Data Lakehouse Infrastructure for Multi-User Access**

Built a Data Lakehouse infrastructure supporting data scientists, government officials, and citizens using Docker, Python, MySQL, MinIO, Adminer, and Apache Superset.

- Implemented data zones for raw ingestion, processing, and access with metadata tracking.
- Enabled SQL querying and table management for government officials via Adminer interface.
- Created interactive dashboards in Superset for citizens to visualize sustainability data.

- **Open Source Contribution: AI-Powered Automatic Subtitle Generator**

Contributed a Python script to automatically generate subtitles (.srt) from audio/video files to an open-source project.

- Used Whisper and FFmpeg.
- Improved accessibility for hearing-impaired viewers and content creators.
- Facilitated transcription of podcasts and lectures using AI-based audio processing.

---

## Technical Skills

**Programming Languages:** Python, Spark

**Libraries / Frameworks:** OpenCV, NumPy, Pandas, scikit-learn, PyTorch, Streamlit, FFmpeg, Whisper, transformers

**Databases:** SQL, NoSQL (postgreSQL, MongoDB...)

**Containerization / DevOps:** Docker

**Version Control:** Git

---

## Languages

**Spanish:** native language

**Chinese:** mother tongue

**English:** C1 - TOEFL 98

---

## Certificates

**AWS Academy Graduate** - AWS Academy Cloud Foundations (2024)

**HackerRank** - Advanced SQL Certificate (2025)

**HackerRank** - Python Certificate (2025)